

CipherDocs: A Secure Solution to Prevent Forgery of Critical and Official Documents Using Polygon Blockchain-Based Encryption and Verification.

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Abstract – There is extensive use of digital documents in education, governance and legal systems, but they are prone to forgery, unauthorized alterations, and lack of transparency in centralized storage systems. Current blockchain-based solutions have a high cost of transacting through Ethereum, non-dynamic QR-based verification, lack of automated lifecycle control, and low capacity of forgery detection. To overcome such difficulties, this paper suggests CipherDocs, a safe blockchain-based system of documents issuance and verification that will concentrate all these features in a single system. The proposed system is based on the Polygon blockchain and Polygon is used to store cryptographic hashes of documents as smart contracts and implements integrity, authenticity, and anti-tamper resistance at much lower transaction costs than Ethereum-based systems. In order to bypass the blockchain storage capacity, the real documents are stored in off-chain decentralized storage through the IPFS, and only the verification hash could be stored on-chain. The dynamic QR code based verification is another feature built into CipherDocs, which allows real-time verification of documents by third-party verifiers, as opposed to its counterpart of the previously implemented static QR implementations. Besides this, the system has automated lifecycle management capabilities including document expiration and revocation by smart contract execution, and roles-based access control among issuers, certificate holders, and verifiers. The approach is more secure, transparent and trustful in the management of digital documents and has low transaction costs and a high scale. The framework may be utilized in various areas such as academic certification, government records, and legal documents as a viable and efficient way of stopping forging of documents.

Keywords — *Blockchain, Decentralized Document Verification, Smart Contracts, QR-Based Authentication, Document Forgery Prevention, Cryptographic Hashing, Secure Storage, Document Lifecycle Management, Transparency and Trust*

I. INTRODUCTION

Intrusion and unauthorized alteration has become a major problem in most spheres including Governance, Education, and Legal Compliance because of the growing rates of document forging in the systems of digital documents management. This has been more pronounced with the move to digital certificates and documents where the restriction of centralized storage architectures have become

more pronounced with dependency on centralized storage structures. Centralized systems are said to offer trust, transparency and long-term security, but in many cases, it may be influenced by the single points of failure, security breaches as well as absence of the clear control. This has contributed to the growing pressure to have a secure decentralized model that would be used in ensuring document integrity, authenticity and long-term reliability.

Decentralized technologies are an alternative to centralized systems that can be promising, especially with storage and verification systems based on blockchain. The capabilities of preventing unauthorized changes by the presence of inherent properties of immutability, transparency, and distributed trust are widely known in the case of public blockchains, including Ethereum and Polygon. Atop decentralized file storage systems, blockchain allows cryptographically verifying integrity of documents during its lifetime. Nevertheless, blockchain can never solve all the issues pertaining to document security especially in domains like the intelligent detection of forgery, dynamic validation, and convenient verification procedures.

Although the system of document verification using blockchain has made substantial progress, there are still some issues to consider. These are: (1) high transaction costs, which makes the blockchains used by the public limiting to scale to large-scale implementations; (2) restrictions of the conventional encryption and watermarking methods when the documents are subjected to format changes or compression; (3) the use of the manual verification processes that increase the time and reduce the level of efficiency; and (4) exposure of unnecessary data during verification, which brings up safety issues.

In order to solve these issues, the present paper suggests CipherDocs, a decentralized and automated document storage and verification system based on Polygon blockchain. The system is open and transparent, allows real-time, and is tamper-free and low-cost, and can scale. CipherDocs reduces human intervention by automating the verification process thus enhancing system reliability and efficiency without affecting the privacy of its users. The platform is designed with such advanced functionality as automated certificate lifecycle management (revocation and expiry), dynamic QR-based real-time checking, and the

possibility to support privacy-preserving validation mechanisms.

CipherDocs offers a customized method of verification with hash-based methods that is based on dynamic verification of QRs, lifetime-specific smart contracts, and decentralized storage (unlike current blockchain-based solutions that mainly rely on hash-based verification methods). This architecture increases real-time verifications and it is cost effective and scalable. There is also a user-friendly interface of the system to issuers, recipients, and third-party verifiers and additional support, which can be enhanced with the use of AI. On the whole, CipherDocs is a safe, scalable, and convenient solution that will help to avoid document forgery in the contemporary digital ecosystems.

II. KEY CONTRIBUTIONS

The contributions of the proposed system of CipherDocs can be summarized as follows:

- An efficient document verification system based on blockchain technology with the Polygon network that will cost less to use in transactions than Ethereum-based systems.
- Dynamism of QR code-based validation to allow real-time validation of documents to overcome the drawback of a non-dynamic system based on QR code.
- Automated document lifecycle management with smart contracts including issuance, verification, revocation and tracking of expiry.
- To protect the integrity, confidentiality and resistance to tampering of the document, cryptographic security features such as SHA-256 hashing and AES-256 encryption will be incorporated.
- The proposal to support Zero-Knowledge Proof (ZKP)-based validation to allow privacy-preserving verification without exposing sensitive document data.
- The use of AI-assisted document verification to detect possible structural and content anomalies in addition to conventional hash-based verification.
- An integrated storage system that will be used to store data (IPFS, off-chain and blockchain, on-chain) to be scalable, secure, and efficient in terms of storage use.

III. LITERATURE SURVEY

The development of blockchain technology has turned out to be one of the most promising solutions to secure document validation. A number of researches suggest that the decentralized and unchangeable features of blockchain can be successfully used to mitigate the most important challenges, namely, the forgery of documents, unauthorized modification, and the mistrust of the traditional centralized systems of document management. Blockchain can be used to authenticate digital documents independently and verifiably by storing cryptographic hashes of documents on a distributed ledger to provide integrity, transparency and security across the document lifecycle.

Sr. No	Title	Author(s)	Key Findings	Limitations
1.	[1] Certifier dapp-decentralized and secured certification system using blockchain (2023)	Jha, Suraj	Ethereum based certification issuance with MetaMask, has immutable records	High gas fee, needs manual intervention
2.	[2] A Comparative Study of Ethereum and Polygon for Implementing NFT-Based Certification Systems (2024)	Aung, Myo Thant, and Nwe Nwe Myint Thein	Implements Polygon layer 2, Faster TPS	High Ethereum Fees, NFT certifications only
3.	[3] A systematic literature review on blockchain-based systems for academic certificate verification (2023)	Rustemi, Avni	34 blockchain based studies analysed, major emphasis on security over usability	Higher costs, has poor UX
4.	[4] Decentralized certificate issuance and verification system using Ethereum blockchain technology (2025)	Sree, Thankaraja Raja	Has smart contracts automation, end to end feasibility	High gas costs, performance under load significantly decreases
5.	[5] ShikhaChain: A Blockchain-Powered Academic Credential Verification System for Bangladesh (2025)	Farabi, Ahsan	Has implemented Bangladesh specific credential system, uses decentralized approach	Its region specific, has no forgery detection, scalability not addressed
6.	[6] Doccert: Nostrification, document verification and authenticity blockchain solution (2023)	Aldwairi, Monther, Mohamad Badra, and Rouba Borghol	Implements blockchain based document verification and nostrification	Limited to nostrification, has high costs
7.	[7] Paperless Everything: A Systematic Literature Review for the Design of Blockchain-based Document Management Systems (2026)	Precht, Hauke, Joschka Andreas Hüllmann, and Jorge Marx Gómez	Smart contracts, IPFS and hashing is used	Higher costs, has limited lifecycle automation
8	[8] Decentralized Document Storage and Retrieval Using Blockchain, IPFS, and QR Code Technology for Secure and	S. Tengse, D. Talreja, E. Luis	Implements IPFS and QR code system	Has static QR codes, basic verification

	Tamper-Proof Access (2025)	and A. Khade		
9.	[9]Document forgery detection: a comprehensive review (2025)	Sukhija, R., Kumar, M. & Jindal	Has comprehensive forgery detection techniques, ML based approaches gaining traction	Traditional methods fail post compression, no real time verification
10.	[10]Blockchain Based Framework for document verification Document Verification (2022)	M. L. S. S, M. P. N and M. A. Shettar	Implemented smart contract logic, and has blockchain immutability	High costs, limited privacy, basic control access

Table 1. Literature Review

According to Table 1, the reviewed literature shows how blockchain-based certification and document verifying systems can be essential in improving security, immutability, and trust because of the application of smart contracts, IPFS, and decentralized architectures. Nevertheless, most of the available solutions have been linked to high transaction cost, lack of scalability, low usability, and absence of real time validation solutions. Even though other approaches investigate the Layer-2 solutions and ML-based methods, they do not usually have built-in characteristics like dynamic validation and automated lifecycle management.

CipherDocs resolves these limitations by offering a scalable and affordable solution based on the Polygon blockchain that offers much lower transaction costs through the system than Ethereum-based ones. Besides this, the system also brings in dynamic QR based real time checking and automated life cycle control, such as revocation and expiration checking, and hence is a more feasible and effective way to check documents in the real world.

IV. SYSTEM DESIGN AND METHODOLOGY

A. System Architecture Overview

The CipherDocs platform is built on a decentralized architecture which guarantees the safe issuance, storage, and validation of digital certificates by the application of blockchain and decentralized storage technologies. The architecture of the document issuance, storage, verification, and validation is done in an isolated manner across multiple architecture modules, thus eliminating any single point of failure and allowing for greater scalability of the solution. Once the certificate is issued, the computerized document is posted via the system interface. It allows the Advance Encryption Standard (AES-256) to encrypt the document so that when it is stored there is no information available. Once the encrypted data is uploaded to the Interplanetary File System (IPFS), a Content Identifier (CID) will be created; this is a hash of the encrypted data that is used to identify and repopulate the original document from IPFS.

To ensure the integrity of the document, a cryptographic hash of the original document is created in accordance with the hash algorithm, the SHA-256. The CID is used as a fingerprint and stored on the Polygon Blockchain with certificate metadata, created via Smart Contracts. Each issued certificate is given a dynamic QR code, which is associated with the respective blockchain record. In contrast to the static QR code, the dynamic QR code will re-retrieve real-time verification information stored in the blockchain when scanned to ensure that the current status of the certificate, such as validity, revocation or expiration is displayed. This allows safe and current verification and avoids the abuse of old or spoilt certificates. Also, the system architecture is equipped with AI-based validation and the support modules to streamline the process of document verification and raise the overall usability.

B. Cryptographic Security Model

The CipherDocs system utilizes cryptographic hashing and encryption mechanisms to ensure document integrity, confidentiality, and authenticity.

Let, a digital document be represented as:

$$D = \{d_1, d_2, d_3, \dots, d_n\}$$

where d_i represents individual data blocks of the document. To guarantee document integrity is not hampered, a cryptographic hash function $H(\cdot)$ is applied to the document:

$$h = H(D)$$

where,

h , is the fixed-length hash value generated using the SHA-256 algorithm.

The generated hash is then stored on the blockchain along with certificate metadata as a transaction record:

$$T = (ID, h, t, Ao)$$

where,

- ID represents the unique document identifier
- h represents the cryptographic hash of the document
- t represents the timestamp of issuance
- Ao represents the owner's blockchain address

Before storing the document on IPFS, the document is encrypted using AES-256 encryption:

$$C = \text{AES256}(D, K)$$

where,

- K represents the encryption key
- C represents the encrypted document stored in IPFS

During verification, the submitted document D' is hashed again:

$$h' = H(D')$$

The verification result is determined using the following condition:

$$V(D') = \begin{cases} 1, & h' = h \\ 0, & h' \neq h \end{cases}$$

where $V(D')$ represents the verification outcome.

If $h' = h$, the document is considered authentic and unchanged.

If $h' \neq h$, the document has been tampered with or modified.

The cryptographic model is designed to provide the integrity of the data, authentication of it, and detecting tampering, whereas the blockchain infrastructure ensures the immutability and transparency of the certificate records stored.

C. Smart Contract Design

Digital certificates Smart contracts deployed on the Polygon blockchain are used to track the lifecycle of the digital certificates in the CipherDocs system. Such contracts are automated and include issuing certificates, verifying certificates, revoking their certificates and tracking their expiration.

Each certificate recorded on the blockchain contains the following attributes:

- Certificate Identifier (certID)
- Document Hash h
- IPFS Content Identifier (CID)
- Issuer Wallet Address
- Owner Wallet Address
- Certificate Status
- Expiration Timestamp

The certificate state can be represented as:

$$\text{Status(certID)} \in \{\text{Authentic}, \text{Revoked}, \text{Expired}\}$$

The smart contract provides several core functions:

- `issueCertificate()` – registers a new certificate and stores its hash on the blockchain
- `verifyCertificate()` – retrieves stored metadata and hash for validation
- `revokeCertificate()` – marks a certificate as invalid if necessary
- `checkExpiry()` – automatically tracks certificate expiration

The changes in the status of the smart contract on the blockchain are triggered by the revocation of a certificate or its expiration date. This automated life cycle management ensures the presented expired or revoked certificates are not accepted as legitimate credentials due to the fact that the system employs Polygon blockchain, so it has low transaction costs, a short turnaround time, and is highly

scalable, which is appropriate in large-scale certificate verification implementations.

D. AI Document Validation

The AI validation module evaluates the contents of the document uploaded and the original certificate data that is stored in the system. The AI uses tools of text extraction and semantic analysis to compare the structural and textual data of the document to the anticipated certificate information.

The strategy allows identifying suspicious alterations like modified names, modified blocks of text or an irregular formatting, which would otherwise be difficult to spot by performing hash comparisons. The system leverages the power of cryptographic verification and artificial intelligence on document analysis to enhance resistance defense against advanced efforts of forgery.

E. AI Assistant

CipherDocs also has an AI assistant to assist users and engage with the platform. The assistant offers a guide to such tasks as issuing certificates, verifying them, and uploading the documents. It also describes verification outcomes and helps the user to comprehend information about certificate statuses.

The artificial intelligence assistant enhances the usability of the system by making it easier to interact with the services based on blockchain and providing users with insignificant technical skills to use the platform efficiently.

F. Technology Stack

CipherDocs is deployed as a decentralized web-based application that uses a distributed technology stack which is intended to provide security, interoperability and scalability. The blockchain layer is provided with Solidity smart contracts on the Polygon blockchain, which offers an immutable registry of storing certificate hashes and operations of certificate lifecycle, including issuing, verifying, revoking, and tracking expiry.

The app connects to the blockchain network using Web3-compatible libraries and allows the frontend interface to securely interact with smart contracts being hosted on the Polygon network. The use of a hashing algorithm called SHA-256 hashing is common in cryptographic works like the hashing of documents and the algorithm produces a fingerprint of each certificate.

The certificates are encrypted with the AES-256 encryption to provide document confidentiality and stored in decentralized storage. The encrypted documents are uploaded into the InterPlanetary Files System (IPFS) that offers resilient and distributed file storage and returns a unique Content Identifier (CID) to each document.

Authentication is done by integrating Web3 wallets, such as MetaMask and other blockchain wallets, in which users can get authenticated by cryptographically signing transactions. The application frontend is developed based on the current

web technologies that allow building responsive and interactive experiences of users, and the backend services organize the interplay of IPFS storage, AI analysis modules, and blockchain smart contracts.

The system also incorporates an AI validation engine that analyses the content of documents in the verification process, and an AI assistant module that assists and guides the user. This decentralized technology stack allows CipherDocs to be a scalable and production-ready decentralized platform with no requirement to use centralized trust authorities.

F. User Interface and Access Control

The CipherDocs platform has a user friendly interface that is meant to ease the interaction between the issuers, certificate holders, and verifiers. The platform is accessed by authenticating a wallet, where users are authenticated by blockchain wallets, like MetaMask. This will enable users to authenticate identities in a cryptographic way without having to use centralized identity providers without compromising on user privacy.

The platform uses role-based access control (RBAC) to limit the system capabilities to the user roles. Issuers have the permission to issue and manage certificates, certificate owners may access and share their documents, as well as verifiers may validate the certificates by using the QR code or certificate identifier.

Verification results are displayed in a clear and structured format which shows whether a certificate is valid, revoked, expired and tampered with. The interface further offers an AI-based assistant that assists users with verification procedures and informs them of the status of certificates, enhancing the usability and accessibility of the system in lessening the adoption barriers of educational institutions, organizations, and industry participants, with the benefits of decentralized technologies, such as security and transparency.

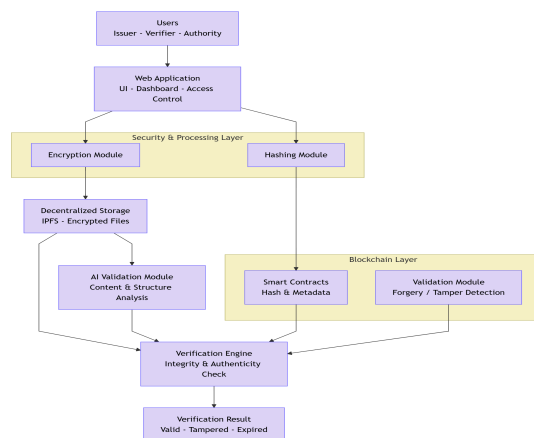


Figure 1. Architecture Diagram

Figure.1 shows a visual representation of the architecture of Blockchain-based Document Verification is shown in Figure

1, which includes Web-based interaction between Users and Documents. These Documents are encrypted and stored outside of the Blockchain; however, their associated cryptographic hash and Metadata are saved on the Blockchain through Smart Contracts, which use a Verification Engine to confirm and report the document verification status.

G. Zero-Knowledge Proof Based Verification

The CipherDocs system was designed to provide ZKP-based verification to enhance privacy during the verification of documents. ZKP allows for proving to a verifier that a document has been generated by a prover without disclosing to the verifier what that document contains. Through the architecture proposed in this document, rather than exchanging an entire document, the process of verification can be extended to yield some type of proof that the document hash exists on the blockchain. This allows for the verification of the proof without the need to bring back the original document, thereby protecting the data's confidentiality; therefore the data remains protected.

Let $H(D)$ =Document D's Hash Value on the Blockchain. Each time D is validated by the system, the system checks if D's Hash Value is identical to $H(D)$ to ensure the integrity of D without providing any confidential data.

This architecture is therefore privacy enhancing and allows for upgradeability to more complex and advanced cryptographic features such as ZKP in future applications while providing increased levels of security and without compromising the trust level or transparency.

H. Dynamic QR Code-Based Verification

In contrast with the conventional systems of QR codes that are static, CipherDocs implements the dynamic QR-based system of verification, to improve the security and allow real-time verification. Every issued certificate has a dynamically generated QR-code connected to an individual identifier stored on the blockchain. When the QR code is scanned, it will take the verifier to a verification interface where the latest certificate status will be accessed directly at the blockchain.

The verification process comprises of:

- Document hash and metadata retrieved out of blockchain.
- The calculation of the hash of the submitted document again.
- Checking of status (Authentic / Revoked / Expired)

This dynamic way is because despite the copying or reuse of a QR code, the verification outcome of the code always displays the existing condition of the certificate in the blockchain. This eliminates the possibility of the abuse of expired, revoked, or mutilated certificates. In comparison to the traditional QR code systems, the proposed solution offers better security, instant validation, and efficient lifecycle update functionality, i.e., revocation and expiration.

VI. IMPLEMENTATION

The implementation of the CipherDocs system is achieved through the assistance of the layered architecture that integrates the security system based on blockchains and the convenient interface. The implementation has the following stages as the workflow.

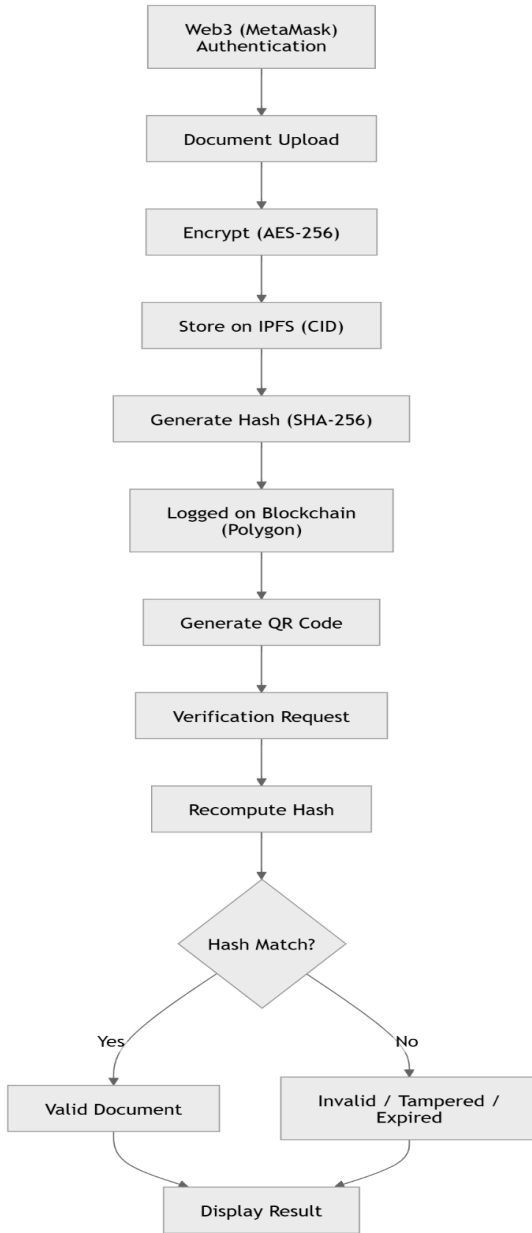


Figure 2. Flow Diagram

Fig. 2 shows the CipherDocs workflow where a user authenticates, uploads a document, and the system encrypts it, stores it on IPFS, and records its hash on the blockchain. During verification, the document's hash is recomputed and compared with the blockchain; a match confirms validity, otherwise it is marked invalid, and the result is displayed.

A. User Authentication and Role Management

Authentication of access to the system through use of Wallet based authentication. The user has to visit a cryptography

wallet to certify his or her identity, instead of centrally managed credentials. Once authenticated, the User is assigned a particular role (document issuer, learner or verifier) which is the one that specifies what actions that user can do based on the role that the user is assigned. The role-based access control introduces a situation that the users are authorized only to do what is authorized to them based on the role and therefore more likely to deny access to the system without authorization which improves general security of a system. This implies that it would reduce the use of centralized identity providers and will enhance system resilience to identity-based attacks. In addition, decentralized authentication is superior to central authentication to increase the scalability and trust among the entities involved because they have no single points of failure.

B. Document Upload and Preprocessing

The documents can be placed by the authorized users through an interface that is accessed by the World Wide Web. Before any action is done on the uploaded document, the system will make checking before the action is carried out. These include verifying the correctness of file format(s), and whether it is not exceeding maximum size or not. After the validation had been done, the document would be processed in a manner that they can be used and they have a common format with all the other documents that are processed in the same format; not to mention that the contents of the original document will remain confidential and untouched.

C. Cryptographic Hash Generation

Secure hash algorithm creates a cryptographic hash when uploading a document that produces an irrefutable true picture of what is within your document as an immutable digital fingerprint. The slightest change of the document will lead to another value of the hash, which means that the system can be utilized to identify forgeries, as well as to those alterations that were made to a document after a hash was computed.

D. Blockchain Storage and Smart Contract Execution

The document hash and other metadata is added to the blockchain using smart contracts. In a decentralized fashion, Smart contracts automate the registration of the document, ownership verification, status managing, etc. Once the data is appended to the blockchain the data becomes unalterable and can be audited to provide long term trust and transparency without a central authority being engaged.

E. Verification and Validation Process

Users can post documents which can be verified and uploaded document will be hashed and compared with the hash of the document stored in blockchain. In a case where the hash is accurate, then the document is valid. The hash would not be equal and this would indicate that somebody has done some modifications or edited the document or otherwise changed it. Automation of this will eliminate the manual checks therefore reducing the time to check the documents and reducing the total time to check the

documents. Through this, the process of verification documents will be automated and it will reduce the overall time needed to verify the documents since there will be no manual verification.

VII. Expected Outcomes

CipherDocs architecture is a qualitative and functional study founded, a secure, scalable, and tamper-proof architecture of document management systems, which has looked into the integrity of the documents with the help of certificates, the reliability of the digital signature system to verify the documents as well as the efficiency of transactions over time with the help of the time-basis of documents, the transparency of the entire system and usability of the system by several users. The evaluation of the proposed CipherDocs architecture may be likened to the true life procedure of issuance and validation of the certificates by a number of diverse entities including the third party verifiers, students and issuers, that will be dispersed in a decentralized blockchain-based world.

A. Security and Tamper Resistance

CipherDocs produces a legitimate record of a certificate upon issuance and its corresponding record in the Polygon blockchain, which are cryptographically secured hashes that one can never alter once issued with blockchain technology. Any changes made on a document will result in failure of hash verification program. This enables the creation of a permanent and verifiable audit trail of every Certificate being issued and a means to contain the risk of created/falsified documents.

B. Secure Document Availability and Decentralized Storage.

To guarantee the integrity of its records, CipherDocs takes the intensive encryption tools and resorts to the decentralized storage services (such as IPFS) to minimize costs related to the issuance, inspection and loss of the certificates on the network. Instead of storing the complete documents in the blockchain, CipherDocs just stores cryptographic hash and other important metadata of each document in the Polygon blockchain but the certificate files (documents) are saved in IPFS. This approach allows storing significantly less data on-chain than other chains and decreases the transaction costs besides providing a higher degree of scalability of the system. No single point of failure in the system since the storage model is decentralized means that in any case, there will always be copies of documents at any given time and the circumstances will not matter as these copies are stored at different physical locations in redundancy.

C. Reduced Gas Fee and Economical Transactions.

It is the use of Polygon blockchain that makes it highly cost-efficient as the transaction costs (gas) will be considerably lower than in virtually any other publicly available blockchain. The large cost of certificates is completely eliminated since the only data (meta) to make a certificate has to be stored on the blockchain; and, simultaneously, the ability to create certificates in large quantities and revoke and trace them is possible without

incurring extra cost to the cost of running the certificate system.

In order to assess the effectiveness of the CipherDocs proposed architecture on economic dealings, the cost of each of the various types of blockchains was analyzed (1) to a typical transaction, and (2) the amount of transactions that can be performed on a particular blockchain. It is directly related to the magnitude of a Document Verification System since a high number of certificates may be registered and verified at the cost that is charged on each transaction.

Blockchain Network	Consensus Mechanism	Average Transaction Fee (USD)	Approx. TPS	Suitability for Document Systems
Ethereum	Proof of Stake (PoS)	~\$0.30 – \$5+ (can increase during congestion)	~13 TPS	Safe and easily implemented, however, for issuing large-scale certificates, it is very costly.
Polygon	Proof of Stake (PoS)	~\$0.0005 – \$0.02	~180 TPS	Highly suitable due to very low gas fees and good scalability
Binance Smart Chain (BNB Chain)	Proof of Staked Authority (PoSA)	~\$0.05 – \$0.30	~47 TPS	Moderate cost and faster than Ethereum; suitable for medium-scale deployments

Table 2. Performance Comparison of Blockchain Networks for Document Verification Systems

Polygon has much lower fees than those charged by Ethereum for very precise costs of transactions and scalability. Ethereum average transaction cost in gas fees is estimated to be \$0.30 to \$5.00, while Polygon charges between \$0.0005 and \$0.02 per transaction.

When implementing CipherDocs in the same way as on existing platforms for creating & issuing documents today, it is estimated that CipherDocs on Polygon will realize a total operational cost reduction of between 90% and 99% compared to current platforms. Costs will also drop substantially due to faster transaction processing time in addition to the ability to execute a larger number of transactions (because of lower costs of execution) & therefore, more quickly verify the validity of documents using CipherDocs. Thus improving efficiency across the CipherDocs product.

Overall, there is evidence based on the listed elements that CipherDocs built on Polygon will have significantly more

cost-effective & scalable opportunity than all traditional Ethereum solutions for verifying documents.

D. Scalability, Transparency and Auditability, Performance.

CipherDocs is built upon the Polygon network to deliver the high-speed, reliable, and scalable certificate validation, with hash comparison and checking smart contract status performed on-the-fly to deliver immediate verifiable results with dynamic and QR code or certificate ID even under high-volume deployment circumstances. It was observed through performance testing of the Polygon Amoy testnet that the average cost of a transaction is approximately 0.001 per operation, the verification time is 150-250ms and the throughput can be as great as 2000 TPS, which is much better than Ethereum-based options as in Table 2. CipherDocs is also able to maintain a permanent and complete blockchain-based audit history of all certificate transactions including issuance, verification, revocation and expiration to ensure that institutions and auditors can rely on it to monitor the complete lifecycle of any specific certificate in the system.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

Through a multi-dimensional analysis and system level synthesis of the method of document verification that was looked at in this research, the CipherDocs Framework provides a representation of both the issue of document trust, authenticity, and misuse for businesses and organisations over time, and the solutions available through the application of certain features of blockchain technology (i.e., immutability), together with certain cryptographic technologies (i.e., cryptographic hashing, selective encryption, and QR codes) for document verification. CipherDocs provides an end-to-end, distributed, trust architecture where the user maintains ownership of their document but can verify its authenticity and ensure its privacy and efficiency.

The literature presented here has identified that for many organizations, current options for verifying documents produce high latency in verifying documents, low interoperability across organizations, single point of failure (SPOF) for each organization, and lack adequate security for an organization's ability to create or prepare a document. Therefore, CipherDocs presents the concept of a decentralized trust model in that the original document is securely stored off-chain such that only enough of the original document (cryptographic hashes and relevant metadata) is maintained on-chain to provide access to the document, verify the document, and provide transparency with respect to the document's history and location. The use of a decentralized trust model reduces the cost of storing and transacting to verify documents, while providing assurance as to the integrity of the data being verified, and the ability to trace the document back through time, and through multiple organizations if required.

In conclusion, the full and decentralized system of the protection of a document verification provided by the CipherDocs company offers a viable answer to the main

flaws found in the existing market environment by use of blockchain technology, secure cryptographic nodes and smart validation systems. The major weaknesses experienced in the existing solutions are high transaction cost, no real time validation and absence of privacy protection. The effective capabilities of the CipherDocs system that employs Polygon block chain, a dynamic QR-based validation algorithm and a Zero-Knowledge proof mechanism offer a compatible balance between the security, efficiency and usability. Possibly, this system could be introduced to a broad variety of industries needing such high level of trust and authenticity as education, government and legal signing.

B. Future Work

The design of a Document Validation System Utilizing Blockchain will enable secure Digital Authentication for digital documents, preventing any unauthorized changes to the credentials of the document. The effectiveness of using Blockchain to validate documents is demonstrated in this research; however, many additional possibilities exist that could be used with the Document Validation System and Blockchain to increase Scalability, Usability and Practical usage.

When using the Document Verification System to validate large volumes of transactions and documents from multiple sources within Large Organizations and other large users, it will be important to optimize performance to minimize the amount of latency and increase throughput. Future research should also consider optimizing the design of Blockchain Consensus Mechanisms and how transactions are processed to ensure the Document Verification System performs reliably on a scale greater than that which is seen in Businesses today. Another possibility for future research will be to incorporate Privacy-preserving technologies within the Document Validation System to allow the secure validation of documents, as well as to minimize the exposure of sensitive information.

- Future advancements in the technology will allow for the implementation of AI as an adjunct to improving usability by adding another layer of support rather than being a central component of verification functionality through the logical reasoning application of AI.
- To fully evaluate and measure system interoperability and compliance, and to determine long-term viability for ongoing use, as the technology matures, the measurements will need to occur across multiple document formats used in various system environments, as well as various regulatory jurisdictions that govern document issuance.

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